

SS Credit: Protect or Restore Habitat

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core and Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1 point)

INTENT

To conserve existing natural areas and restore damaged areas to provide habitat and promote biodiversity.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Preserve and protect from all development and construction activity 40% of the *greenfield* area on the site (if such areas exist).

AND

Restore a portion of the site, including the building footprint, identified as *previously disturbed (if such areas exist)* and follow vegetation and soil requirements below. Vegetated roof surfaces may be included in the habitat area calculations if the plants are native or adapted and provide habitat. Points are awarded according to Table 1.

Table 1. Points for percentage of area restored

Restored Area	Points BD+C (except Healthcare)	Points BD+C (Healthcare)
15% of previously disturbed area	1	1
25% of previously disturbed area	2	-

Soil Restoration

Restore disturbed soils in areas that will later serve as the final habitat area.

- ▶ Imported soils may not include the following:
 - soils defined regionally by the Natural Resources Conservation Service web soil survey (or local equivalent for projects outside the U.S.) as *prime farmland*, unique farmland, or farmland of statewide or local importance;
 - soils from other *greenfield* sites; or
 - sphagnum peat moss

Vegetation

Plant a minimum of 6 species of vegetation that are native or adapted to the project's EPA Level III ecoregion (or local equivalent for projects outside of the U.S.). Include a minimum of 2 out of the following plant categories: tree, shrub, and ground cover. Designate a portion of the habitat area for a pollinator garden consisting of native flowering plants and totaling at least 30 square feet (3 square meters).

Schools only:

Dedicated athletic fields that are solely for athletic uses are exempted from counting toward the total site area. These areas may not count toward the protected greenfield or restored habitat areas.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit is now more achievable for various project types and more directly addresses the intent of the credit as it relates to habitat protection and revitalization. The restoration requirement was reduced from a minimum of 30% to 15%, and a second threshold was added at 25% for two points to reward projects for increased restoration. Soil requirements have been streamlined and testing requirements have been removed. A vegetation section was added to encourage planting of diverse vegetation and pollinator habitat. Option 2. Financial Support was removed to encourage focus on local habitat restoration.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modifications:

Verify that at least 15%, rather than 30%, of previously disturbed areas will be restored.

- ▶ Soil tests are no longer required.
- ▶ Option 2 Financial Support is no longer offered in this credit.
- ▶ Vegetation:
 - A contiguous series of habitat areas, including pollinator gardens, connected by landscaped paths are encouraged in order to facilitate connectivity for wildlife, especially pollinating insects that cannot travel far without access to food sources. Pollinator gardens are recommended for areas such as parking lots, which include several small contiguous parcels of vegetation.
 - *Ground cover* includes native and adapted grasses as well as perennials and other plant species. See WE Credit Outdoor Water Use Reduction for more guidance on ground cover.
 - Refer to the Xerces Society's Pollinator-Friendly Plant Lists as a resource to select appropriate plants for local pollinators.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Best Practices for Habitat Protection:
 - For more information on invasive vegetation, refer to the USDA's NRCS PLANTS Database, the Invasive Plant Atlas of the United States website, and/or state and federal Noxious Weed laws (or local equivalent for projects outside of the U.S.).
- ▶ Vegetated Roofs:
 - Projects no longer need to meet a floor-area ratio (FAR) density minimum for vegetated roofs to be considered part of the restored area.

Required Documentation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Site plan depicting the project boundary, building footprint(s), preserved greenfield area(s) (if applicable), previously disturbed area, restored habitat area, native and adapted vegetation, and designated pollinator garden.

- ▶ Plants list, including pollinator plants, appropriate for project's EPA Level III ecoregion (or local equivalent for projects outside the U.S.); must include a minimum of 2 out of the following plant categories: tree, shrub, and ground cover.
- ▶ Description of disturbed or compacted soils to be revegetated
- ▶ EPA Level III Ecoregion description (or local equivalent)
- ▶ No longer required: Soil test results.
- ▶ No longer required: Projects with vegetated roofs to provide the floor area ratio (FAR)
- ▶ Option 2: Financial Support has been removed and relevant documentation requirements are no longer applicable.

Referenced Standards

Refer to the LEED v4 reference guide, with the following additions:

- ▶ **EPA Level III Ecoregion Descriptions:**
ftp://newftp.epa.gov/EPADDataCommons/ORD/Ecoregions/pubs/NA_TerrestrialEcoregionsLevel3_Final-2June11_CEC.pdf
- ▶ **USDA Natural Resources Conservation Service, PLANTS Database:**
<https://www.nrcs.usda.gov/wps/portal/nrcs/main/national/plantsanimals/plants/>
- ▶ **Invasive Plant Atlas of the United States:** <https://www.invasiveplantatlas.org/>
- ▶ **U.S. State and Federal Noxious Weed laws:** <https://plants.usda.gov/java/noxComposite>
- ▶ **Xerces Society for Invertebrate Conservation, Pollinator-Friendly Plant Lists:**
xerces.org/pollinator-conservation/plant-lists/

Exemplary Performance

- ▶ Restore a minimum of 35% (including the building footprint) of all previously disturbed area on site using soil and vegetation that meet the credit requirements.

Definitions

- ▶ **adapted vegetation** vegetation that is not native to the particular region it was introduced to, but has evolved or maintained characteristics conducive for healthy growth and requires no additional resources or maintenance, such as water for irrigation, in comparison to similar species native to the area. An adapted species is non-aggressive or disruptive to native plant communities.
- ▶ **contiguous** adjoined or adjacent parcels that facilitate connectivity
- ▶ **greenfield site** area that has not been graded, compacted, cleared, or disturbed and that supports (or could support) open space, habitat, or natural hydrology. See also: previously disturbed.
- ▶ **habitat area** area that supports native wildlife populations by providing locations for nesting, cover, respite, and/or food sources.
- ▶ **invasive vegetation** nonnative or exotic vegetation that quickly grows and aggressively reproduces, often disrupting or displacing native plant communities. The plant's vigor combined with a lack of natural enemies leads to outbreak populations.
- ▶ **pollinator** an insect or animal that moves pollen from the male anther of a flower to the female stigma of a flower in order to pollinate the vegetation.
- ▶ **pollinator garden** area that is planted with native flowering vegetation that provide nectar, pollen, habitat, or nesting area for native pollinating fauna.
- ▶ **previously disturbed** areas that have been graded, compacted, cleared, previously developed, or disturbed in any way. These are areas that do not qualify as 'greenfield.'

Connection to Ongoing Performance:

- ▶ LEED O+M credit Site Management: Naturally vegetated areas that meet the requirements of this credit are less likely to require the routine use of maintenance equipment. They could also reduce irrigation, nutrient application, and erosion, thereby helping the project meet the requirements of the related performance-based credit.
- ▶ LEED O+M SS credit Rainwater Management: Restoring and protecting greenfield areas with native or adapted vegetation will help provide applicable strategies to earn the related performance-based credit.
- ▶ LEED O+M SS credit Heat Island Reduction: Vegetated roofs can be counted for both credits.